

Opioid Overdose on the Ward

Section 1: Case Summary

Scenario Title:	Opioid Overdose on the Ward
Keywords:	Opioids, acute care medicine, airway
Brief Description of Case:	This case involves the identification and initial management of the opiate intoxicated patient. The resident is expected to initiate appropriate supportive measures (ensure working IV, oxygen, monitors) while obtaining a focused history and performing a focused physical exam. They should call for help (RACE, anesthesia, ICU) due to decreased LOC and hypoxia.

Goals and Objectives	
Educational Goal:	To practice the initial assessment and management of an acute opioid overdose
Objectives: (Medical and CRM)	1. Review an approach to the patient with decreased LOC 2. Prioritize initial investigations and resuscitation therapies 3. Begin basic airway intervention 4. Call for help (RACE, anesthesia, ICU, etc).
EPAs Assessed:	

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners		☐ Senior Learners	
	☐ Staff			
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab		☐ In Situ	
	☐ Other:			
Recommended Number of Facilitators:	Instructors: 1			
	Sim Actors: 1			
	Sim Techs: 1			

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
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Revised By:	Dr Chris Heyd
Version Number:	



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Joseph Lewis		Age: 42		Gender: M	Weight: 80kg
Presenting complaint: Decreased level of consciousness					
Temp: 36.6 °C	HR: 102	BP: 125/60	RR: 8	O ₂ Sat: 82%	FiO ₂ : Room Air
Cap glucose: 8.8			GCS: 9 (E2 V3 M4)		
Triage note: It's 2300 and you're called to assess a 42M who has been admitted to the orthopedics service for a right tibial osteomyelitis. He has had an initial debridement and is awaiting a second surgery in 48 hours. His RN is concerned as he was difficult to rouse on evening rounds.					
Allergies: None					
Past Medical History: Type 2 diabetes Rheumatoid arthritis Smoker			Current Medications: Metformin 500mg PO BID Celebrex 200mg PO BID Cyclosporine 75mg PO BID Pip/taz 3.375g IV Q6H Morphine via PCA and PRN oral morphine PRN tylenol, zopiclone, gravol		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
RN to provide when asked: You've been looking after Mr. Lewis. He was quite uncomfortable (leg pain) at the start of your shift and required 2 doses of 10mg oral morphine (at 1900 and 2200) as well as his PCA doses of IV morphine. On your subsequent vitals you find him somnolent and difficult to rouse.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal	Neuro: Single inappropriate words, pulls away from pressure, pupils 2mm
Resp: Slow, shallow breathing	Head & Neck: Perioral cyanosis
Abdo: Normal	MSK/skin: Bandaged leg, otherwise nil
Other: Normal	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin: Adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
PCA pump Airway equipment (BVM, oral airway, nasal airway) Portable vitals machine
C. Required Medications
Naloxone
D. Moulage
Hospital gown, right lower leg in bandage
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Perioral cyanosis Constricted pupils Slow, shallow breathing Difficult to rouse, sleepy inappropriate single-word answers

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Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside RN	RN notes: • You've been looking after Mr. Lewis. He was quite uncomfortable (leg pain) at the start of your shift and required 2 doses of 10mg oral morphine (at 1900 and 2200) as well as his PCA doses of IV morphine. On your subsequent vitals you find him somnolent and difficult to rouse. • Wait for the resident to ask for vitals, O2, etc. • IF the resident doesn't appreciate the low respiratory rate, mention "he's breathing really slowly... I don't think he's breathing".

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tach HR: 102 BP: 125/60 RR: 8 O ₂ SAT: 82% T: 36.6°C GCS: 9(E2 V3 M4)	<i>Sonorous with sounds of obstructive upper airway</i>	<u>Expected Learner Actions</u> ☐ Monitors ☐ CBG ☐ Focused history and physical ☐ Place patient on oxygen ☐ Jaw thrust, chin lift ☐ Ask for crash cart including airway equipment ☐ Call for help (RACE/ICU) ☐ Ask for naloxone	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> O ₂ given → no change <u>Triggers</u> <i>For progression to next state</i> 4 minutes or all actions complete → 2. Hypopnea	
2. Hypopnea Rhythm: Sinus tach HR: 108 BP: 125/60 RR: 6 O ₂ SAT: 82%	<i>RN can prompt: "he's breathing really slowly... I don't think he's breathing"</i> <i>Ongoing sonorous respirations</i>	<u>Expected Learner Actions</u> ☐ Place oral or nasal airway ☐ Ventilation by BVM ☐ Call for help (RACE/ICU) ☐ Ask for naloxone	<u>Modifiers</u> BVM → sat 90% BVM+ airway → sat 94% No ventilation by 5 min → sat 75% <u>Triggers</u> - All actions complete → 3. Assisted Vent	BVM arrives at beginning of state
3. Assisted Vent Rhythm: Sinus HR: 90 BP: 125/60 RR: 12 (assisted) O ₂ SAT: 94% GCS: 9(E2 V3 M4)		<u>Expected Learner Actions</u> ☐ Give naloxone IV ☐ Discontinue PCA pump ☐ Continue airway support ☐ Continue monitoring ☐ Reassess after naloxone ☐ Discussion with RACE/ICU	<u>Modifiers</u> - naloxone → RR12, GCS12 <u>Triggers</u> - Transfer discussed with RACE/ICU → END CASE	Naloxone arrives at beginning of state



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Appendix A: Laboratory Results (From AM Bloodwork)

CBC WBC 14 Hgb 130 Plt 220 Lytes Na 137 K 3.8 Cl 97 HCO₃ 24 Urea 7.4 Cr 137	
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

No images used in this case.

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. Review the initial approach to the patient with decreased LOC
2. Discuss the approach to the patient with opiate intoxication
 - Naloxone (competitive opioid antagonist)
 - 0.4-2mg q2-3mins for ACUTE overdoses. May use up to 10mg. Titrate to reversal of respiratory depression. Once achieved then infusion at “reversal dose”/hour. How do you treat acute pain once naloxone has been given?
 - Much smaller in doses for patients on chronic opiates – 0.1mg. May potentiate severe withdrawal if given too aggressively in this population.
 - Patient must be monitored (may be started on the floor with RACE team but plan to transfer to a monitored bed)... The half-life of naloxone is much shorter than most opiates, so recurrent respiratory depression may occur.
3. Review the skill of basic airway maneuvers to improve oxygenation (establish an airway) and ventilation (BVM).

References

1. <https://emergencymedicinecases.com/opioid-misuse-overdose-withdrawal/>
- 2.
- 3.

